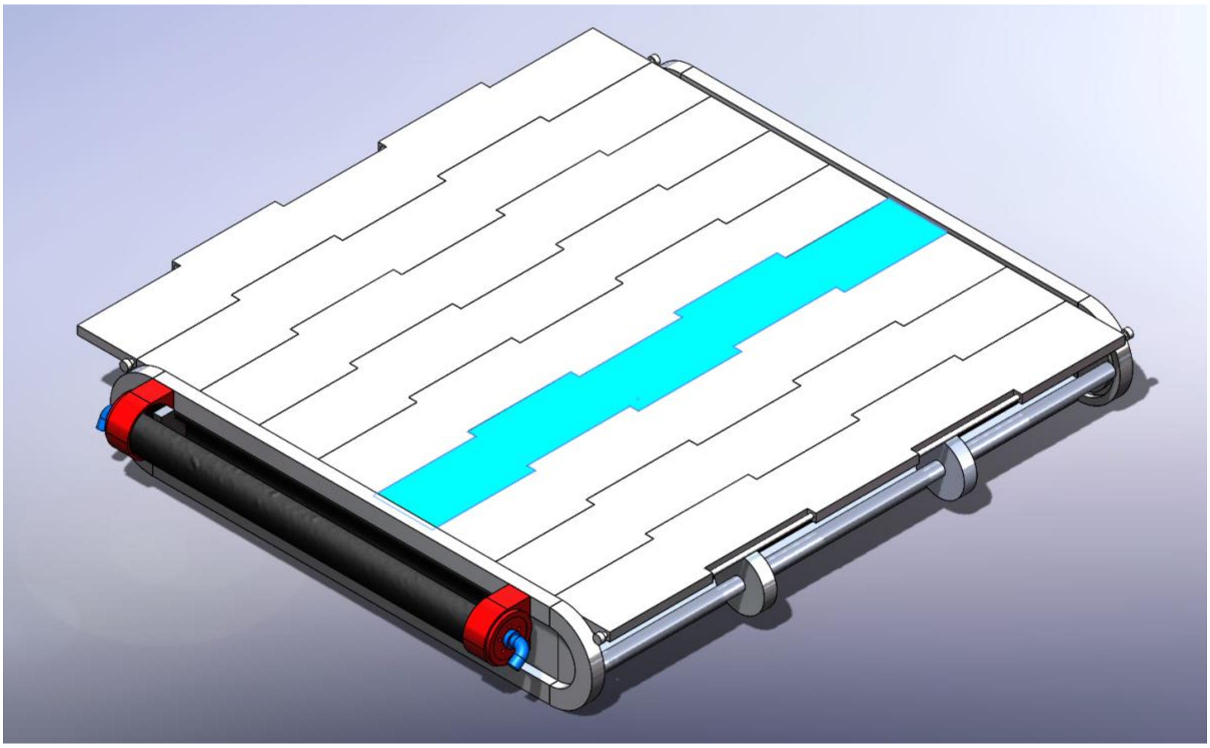
ARTEFACT 3

DESIGN CRITIQUE



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UNIVERSTY OF TECHNOLOGY SYDNEY | FACULTY OF ENGINEERING AND INFORMATION TECHNOLOGY

41066 MECHANICAL SYSTEMS DESIGN STUDIO

GROUP 25

TABLE OF CONTENTS

1 ACTUATION SYSTEM REDESIGN

1.1 IDEATION (NEO)

For the efficiency of the actuation system, the radiator will be reorientated to limit the distance of travel required by the system. It will be turned 90 degrees, with the longer side parallel to the louvre blades, reducing the length from 610mm to 470mm.

In the Ideation phase, our group gathered for a brainstorming session where each member would consider as many design options as possible for each subsystem within a 30-minute time frame and the results would then be compiled. After compiling the design options, we would together clear out any options that were not feasible or realistic, leaving us with a refined morphological table. Scoring matrices were then used to rank the options by criteria of: Cost, Reliability, Ability to (e.g. to extend), Weight, Size. The best solutions were then developed further with prototyping iterations.

Requirements:

- Ability to Extend 470mm
- Ability to provide enough force to push blades (required force)
- Low cost
- High Reliability
- Low complexity
- Low Weight
- Small Size (fit within 70mm height requirement)

Different versions were modeled to test our scoring of Ability to extend, with each configuration showing pros and cons.

1.1.1 BIMETALLIC COIL (NEO)

A continuation from the previous sprint's design, a bimetallic coil attached to a pulley system was prototyped to consider functional compatibility. After implementing and furthering calculations for the coil's length, material and power, it was concluded that this would be a suboptimal method of actuating the louver blades as it requires (insert calculations of system)

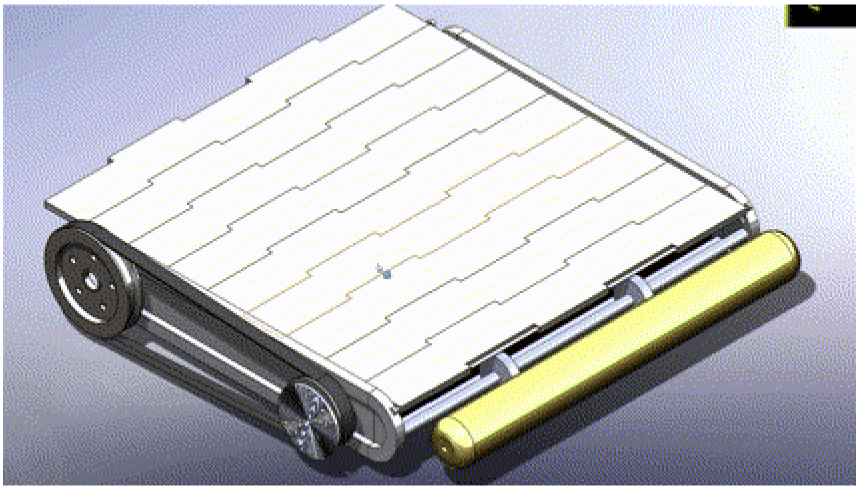


Figure 1Pulley concept

1.1.2 GAS EXPANSION TELESCOPIC PISTON (NEO)

After brainstorming, an idea that stood out was the concept of a telescopic piston that would self-actuate due to expanding gases within its internal cylinder. This concept would be passively powered, provide enough force to push the blades and would be relatively compact, fitting snugly beside one of the side frames. However, calculations for the piston's required volume to extend by the required 470mm showed an extreme 12.5L of capacity required (show calcs in appendix). This would mean an external tank of that capacity would be required for the design to function, counteracting the positive size qualities of this design by needing a tank exceeding the maximum height requirement of the system. An alternative solution was prompted by this development.

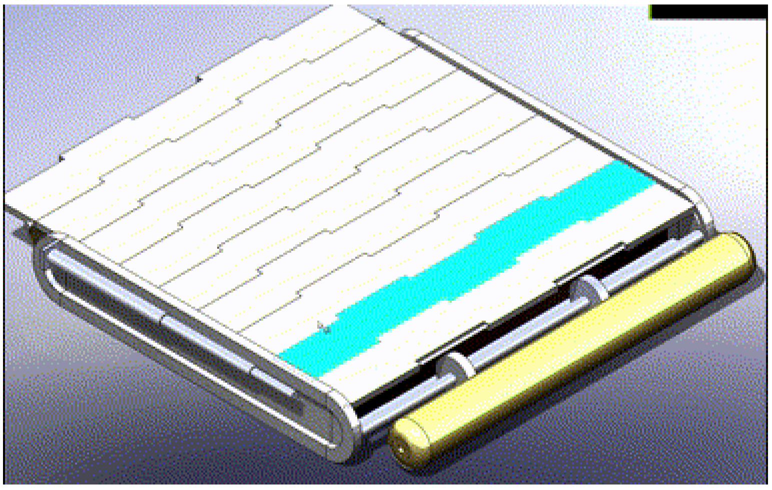
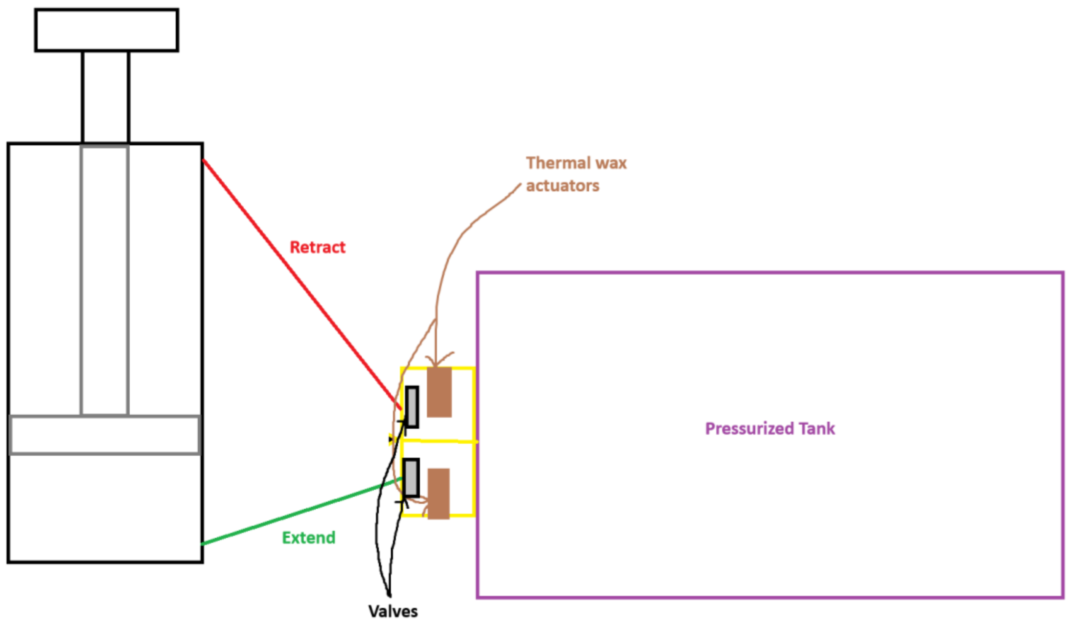


Figure 2 Telescopic piston concept

1.1.3 PRESSURIZED TANK POWERED PISTON (NEO)

Building from the knowledge that an external tank would be required for a piston system, a regular linear actuator piston powered by pressurized tank was explored. The initial framework for the system follows an almost identical framework as a regular pneumatic piston using a compressor with an inlet at each end of the piston cylinder pushing the piston rod forward or backward. The direction would be controlled by valves that bias airflow between the two air inlets. The adjustments made to this system to meet the passively powered requirement include a tank that constantly applies a pressure through the air hoses, as well as the valves being controlled passively by thermal wax actuators, opening and closing depending on the temperature of the radiator.



This concept showed a lot more promise compared to previous ideas, but further maturation of the design showed major shortfalls in:

- Requiring the ability and a method to release pressure from the cylinder to allow for movement in opposite direction
- The mounting orientation of the piston being far from optimal (demonstrated in the CAD model)
- Lacking a way for valves to be opened and closed at specific temperature windows

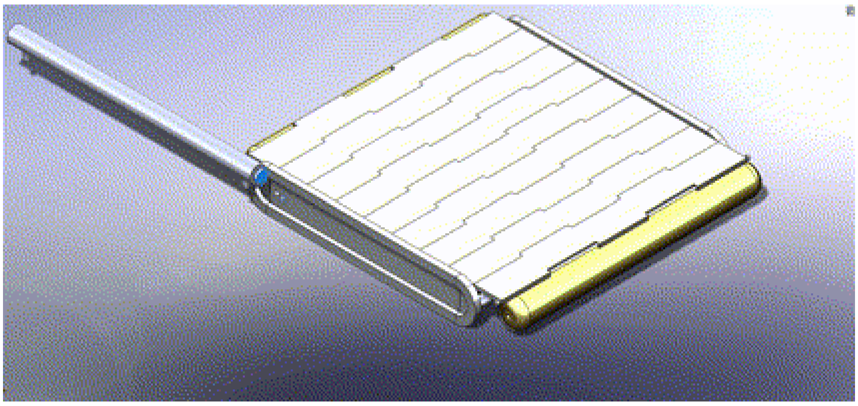


Figure 3 Piston concept

1.1.4 INFLATABLE ACTUATOR CONCEPT (NEO AND DARCY)

Drawing inspiration from inflatable tube puppets, the concept of an inflatable tubes that could expand and contract allows to overcome the length and mounting issue that the previous piston idea possessed. This concept would simplify and minimize the footprint of the entire louver system whilst maintaining high ability to extend and functionality. Further development of this concept was undertaken to create a fully working prototype.

- What thermal wax actuator to be used?
- Regulator to be added to limit expelled gases
- Way to control release valve to be developed. (darcy's idea of a sliding blocking valve)

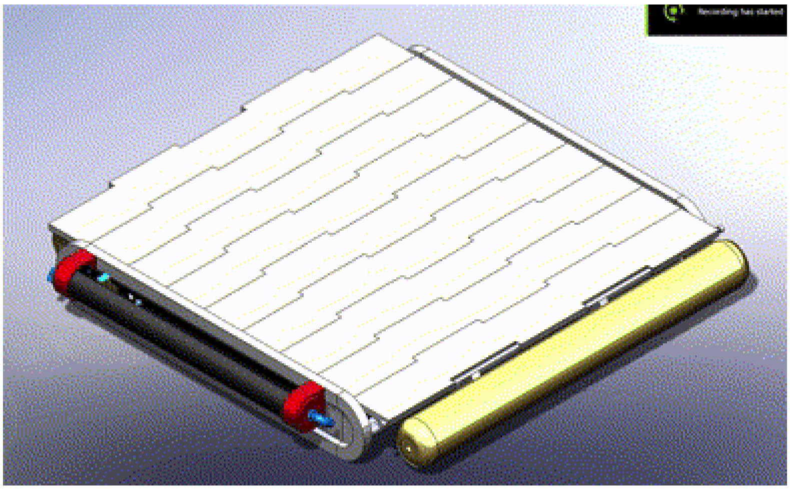


Figure 4 Inflatable actuator concept

2 FINAL PROTOTYPE DESIGN FUNCTIONS

2.1 PRESSURIZED TANKS

2.1.1 GAS OPTION

Due to the extreme temperatures and conditions of outer space, the choice of gas used within the tanks is a vital decision that could lead to catastrophic failure if not done carefully.

Selection Criteria

Temperature Compatibility: Must be inert, non-flammable and operate at a wide temperature range. The gas remains stable and does not condense, freeze, or react under operating temperature conditions.

System Compatibility: The gas should be compatible with the materials of the pistons, valves, and actuators to avoid corrosion or chemical reactions.

Cost and Availability: Consider the availability and cost-effectiveness of the gas, especially if it needs to be refilled frequently.

The options for consideration included Nitrogen, Helium, Carbon Dioxide and Compressed air.

Nitrogen (N_2):	**Application Suitability:** A stable, inert gas suitable for a wide range of thermal conditions. **Advantages:** Widely available, cost-effective, and compatible with most materials in engineering systems.
Helium (He):	**Application Suitability**: A lightweight, inert gas that maintains stable pressures across varying temperatures. **Disadvantages:** Higher cost compared to other gases, making it ideal only when its low weight and specific properties are critical.
Carbon Dioxide (CO_2):	**Application Suitability:** A non-flammable gas effective for lower-pressure systems and easy to compress. **Considerations:** Potential for dry ice formation at extremely low temperatures, which may impact system performance.
Compressed Air:	**Application Suitability:** A practical and economical option if the system includes proper filtration to manage moisture. **Considerations:** Must be filtered and dried to prevent contamination and condensation, especially in low-temperature environments.

Safety: The gas's safety in terms of flammability, toxicity, and chemical stability.
Thermal Stability: The ability of the gas to perform under a wide range of temperatures.
Availability: How easily the gas can be sourced and filled for the application.
Cost: The economic feasibility of using the gas, considering both initial and operating costs.
System Compatibility: The compatibility of the gas with the system's materials and components.
Weight Efficiency: The gas's effectiveness in maintaining pressure without adding significant weight.

Gas Type	Safety (1-5)	Thermal Stability (1-5)	Availability (1-5)	Cost (1-5)	System Compatibility (1-5)	Light Weight (1-5)	Total Score
Nitrogen (N_2)	5	5	5	5	5	4	29
Carbon Dioxide (CO_2)	4	4	5	5	4	4	26
Compressed Air	5	4	5	5	5	3	27
Helium (He)	5	5	3	2	5	5	25

After creating a scoring matrix to determine the best fit, **Nitrogen** scored the highest due to its safety, excellent thermal stability, wide availability, and cost-effectiveness. Its high compatibility with most materials and components also makes it an ideal choice. The only slight drawback is that it isn't as light as helium, but the difference will be negligible for this application.

2.1.2 PRESSURE

Due to the vacuum of space, the surrounding pressure around the louvre system will be equivalent to zero Pascals.

2.1.3 VOLUME

Considering the pressure required for actuation and the number of cycles required for each mission, the volume of the pressurized tanks was

2.1.4 SHAPE AND CONFIGURATION

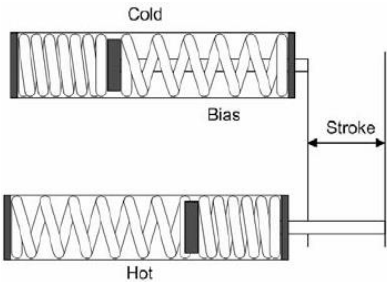
To optimize performance of the pressurized gas and for safety reasons, an insulated chamber to house the tanks will be implemented. The insulation

2.1.5 REGULATOR

2.2 VALVE ACTUATION (DARCY)

Thermal Actuator Selection

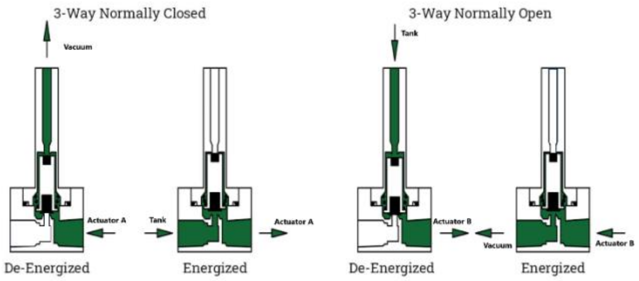
- Rubitherm RT21HC
    - Melting/Solidification point of ~21°C
    - Volume Expansion ~14%
    - Max Working Temp 45°C
- Rubitherm SP21EK
    - Melting/Solidification point of ~21°C
    - Volume Expansion ~6%
    - Max Working Temp 45°C
    - Corrosive to metals
- Two Way Shape Memory Alloy Spring



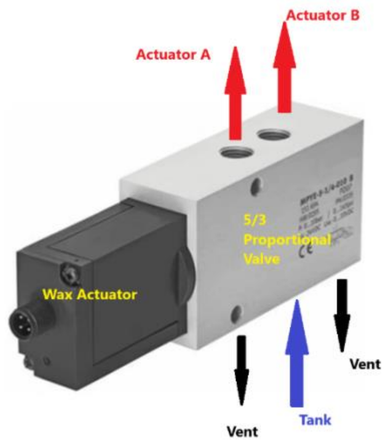
- Would need a specific design to have correct stroke and temperature response.
- Very reliable
- Greater operational temperatures

Valve Selection

- Custom Valve
  - Initial Ideation
  - Does not give full control
  - Reliable
- 2 3-Way Solenoids



  - Proven design
  - Does not give full control
  - Reliable
- 5-Port 3-Position Proportional Valve



  - Commonly used in double action hydraulic cylinders, which this novel inflatable design takes inspiration from.
  - Allows pressure control to give full range of control over shutter movement
  - Could be paired with linear actuator to control valve movement rather than typical electronic control

- Proven Design
- Reliable

3 REDESIGN MASS, SIZE & THERMALS (BEN)

3.1 SIZE (BEN)

Having determined a redesign of the given louvre system the redesigned louvre system demonstrated in the form of inflatable actuator design accommodates precisely the 450x600mm radiator for this mission at a height of 70mm, as stipulated at the beginning of the project. Just as Neo initially mentioned, a transformation of the dimensions of the given louvre system was performed to shorten the stroke length of the actuator while allowing for an increase in the length of the gas tanks along the longest side of the louvre system. As a result, the louvre blades were lengthened to extend across the now 630mm gap. This development provides a significant advantage because the friction to overcome of the wheels to displace the louvre blades remains the same regardless of the configuration, and because the gas is consumed from purging to space when returning the actuator to rest more gas from a bigger tanks means more actuation cycles for a longer mission time.

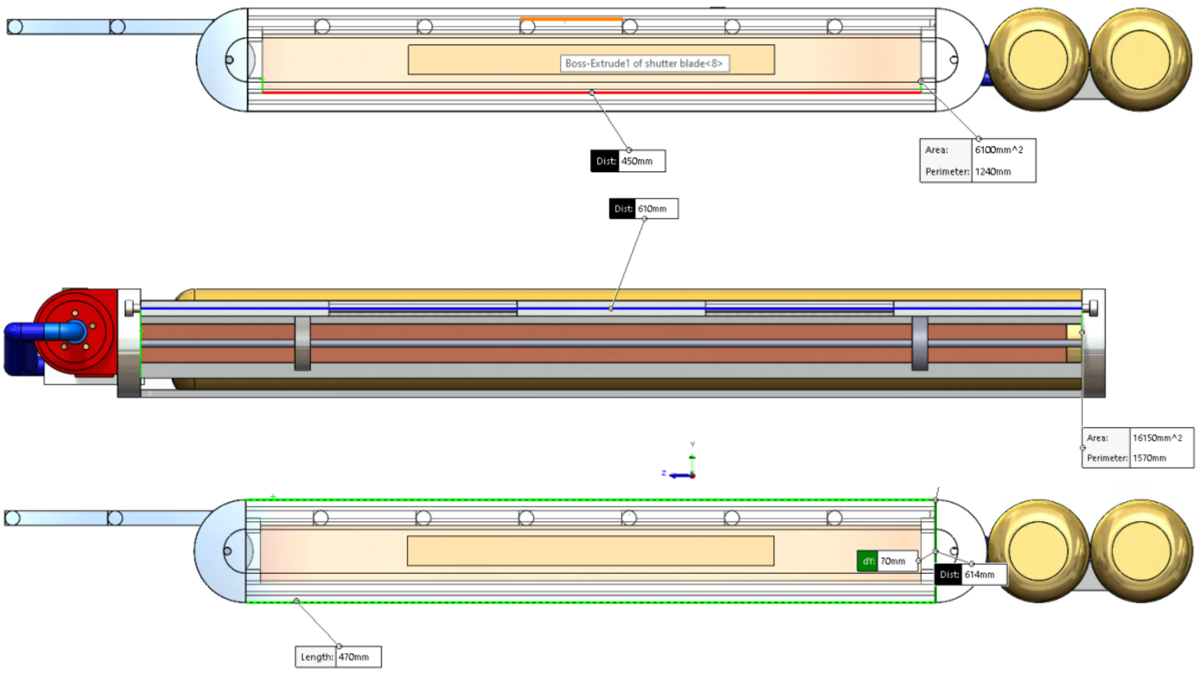


Figure 5 - Dimensions of inflatable actuation louvre system about radiator cavity.

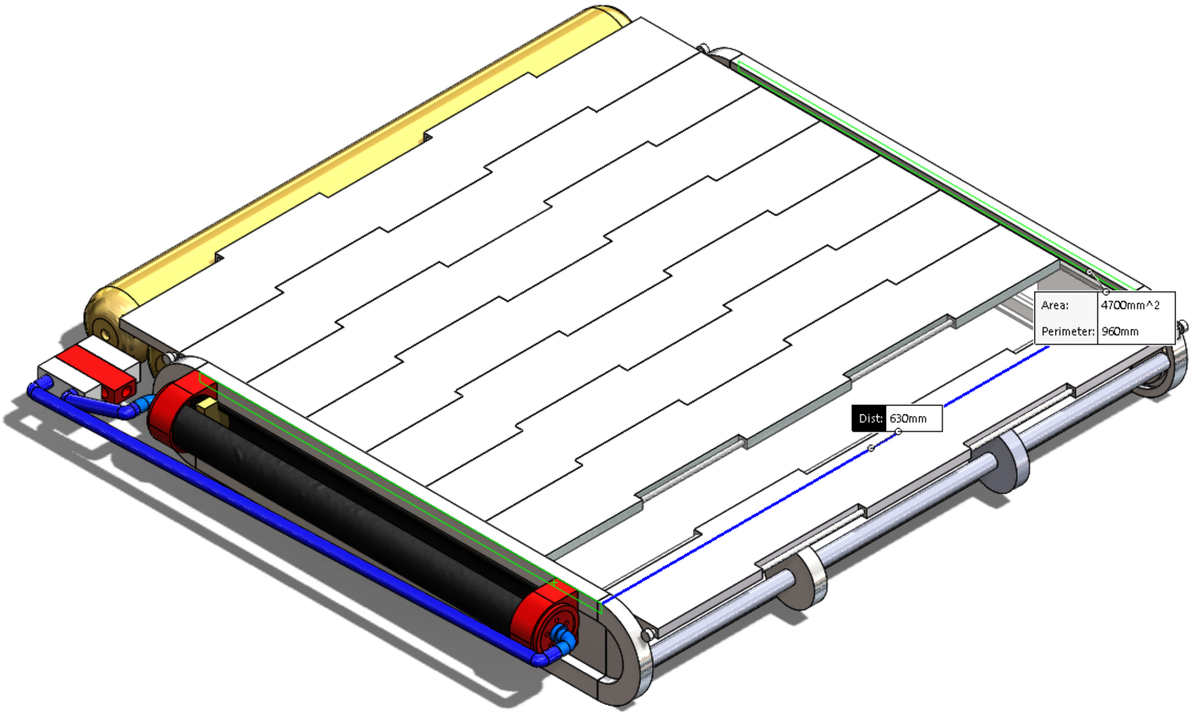


Figure 6 – Extension of louvre blades.

3.2 MASS (BEN)

In reducing and transforming the dimensions of the given louvre, the total surface area of the redesign is 2.774m^2, and the total mass implementing 6063-T4 roughened aluminium throughout to ensure a heat rejection ratio of ten times between open and closed states as discussed in the previous sprint was determined to be 18.072kg. This resolves a weight to area ratio of 6.515kg/m^2 for the redesigned louver system. This fulfils the functional requirement for the mass of this redesigned louvre as it is lower than the upper spec of the Starsys louvre assembly flown in past missions.

Table 9.1. Characteristics of Flight-Qualified Rectangular-Blade Louver Assemblies[a]

	OSC	Swales	Starsys
Blades	3 to 42		1 to 16
Open set points (°C)		0 to 40	−20 to 50
Open/close differential (°C)	10 or 18	10 or 18	14
Dimensions (cm)			
Length	20 to 110	27 to 80	8 to 43
Width	36 to 61	30 to 60	22 to 40
Height		6.4	6.4
Area (m^2)	0.07 to 0.6	0.08 to 0.5	0.02 to 0.2
Weight/area (kg/m^2)[b]	3.2 to 5.4	~ 4.5	5.2 to 11.6
Flight history	Nimbus, Landsat, OAO, ATS-6, Viking, GPS, SolarMax, AMPTE, SPARTAN, Hubble, Magellan, GRO, UARS, EUVE, TOPEX, GOES, MGS, MSP	XTE, Stardust	Rosetta[c], Quickbird[c] JPL:[d] Mariner, Viking, Voyager, Galileo, MLS, Magellan, TOPEX, NSCAT, Cassini, Seawinds

[a]This table contains representative values from past louver designs. Contact manufacturer for additional design possibilities or values for specific designs.
[b]Weight without sunshield.

Table 1 – Characteristics of Flight-Qualified Rectangular-Blade Louver Assemblies.

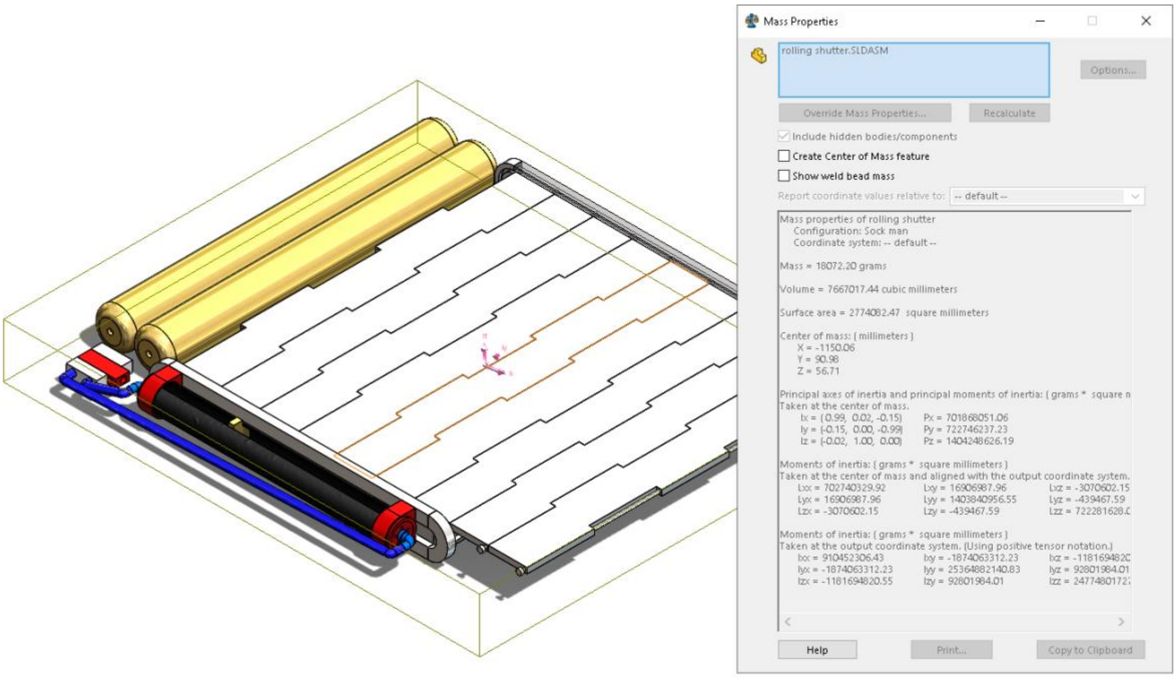


Figure 7 - Mass properties of inflatable actuation louvre system.

3.3 THERMALS (BEN)

In this redesign the louvre still operates in a manner where it can either cover the radiator completely or not at all. This results in effective emissivity being the same as the radiator and louver materials at 100% opened and closed respectively. Therefore, the fully opened state is simply the emissivity of the radiator, and the fully closed state is the emissivity of the louver.

Roughened aluminium with an emissivity of 0.07 was eventually settled on to push the heat rejection capacity right over the edge of the 10 times ratio from 629.9W opened to 50.1W closed to fulfil the thermal functional requirement.

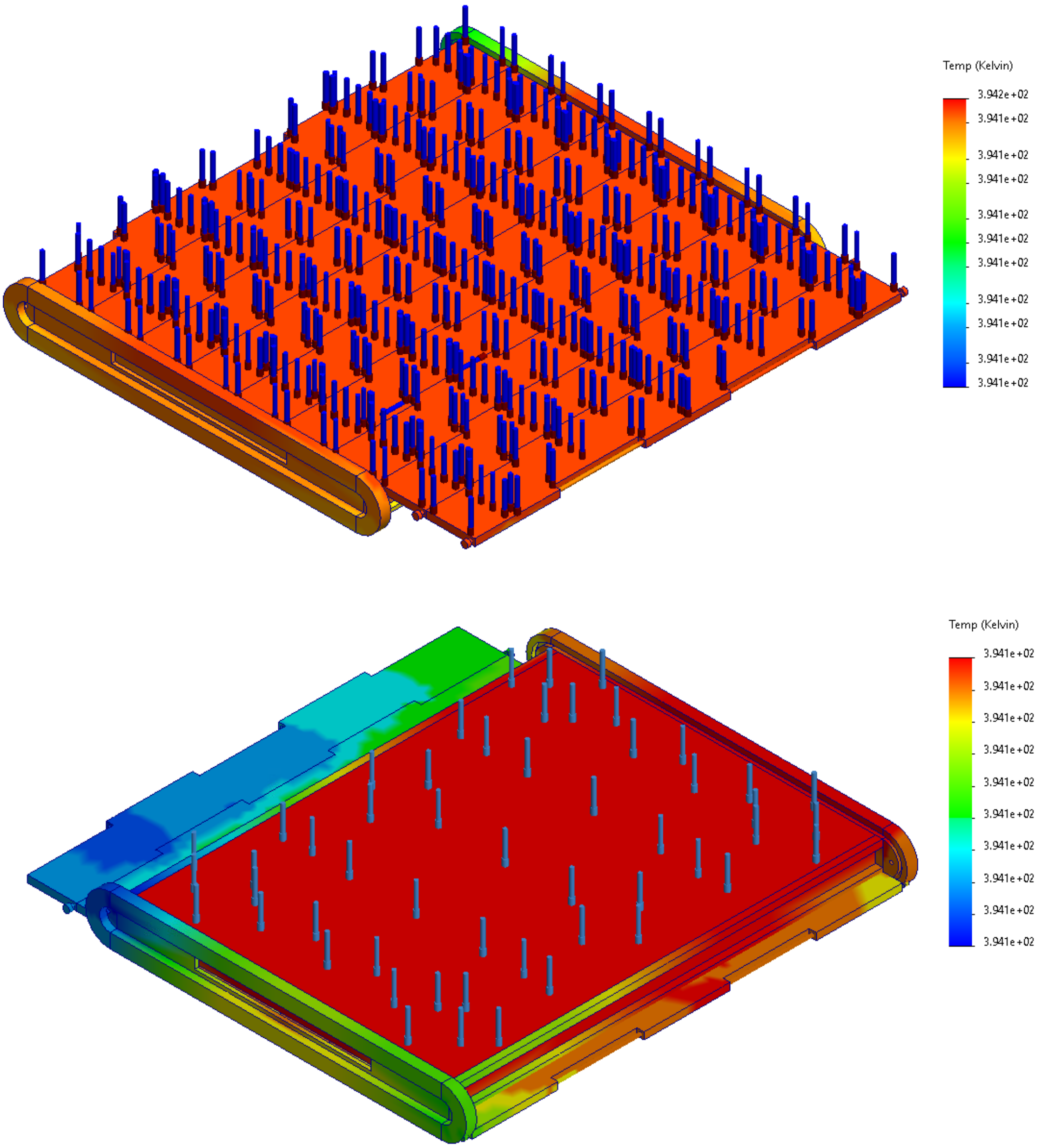


Figure 7 - Thermal loading of redesigned louvre.

4 COST ANALYSIS (NHAN)

4.1 PROJECT PHASES FOR LOUVER SUBSYSTEM

Initial Assessment and Planning: 2 months
Design and Prototyping: 6 months
Testing and Validation: 6 months
Final Design and Implementation: 5 months
Integration and Launch Preparation: 5 months
Total Estimated Duration: 2 years
The project would take approximately 2 fiscal years to complete. This estimate is more aligned with the development of the louver subsystem for LEO satellites

4.2 PERSONNEL ON THE TEAM

Scientist Personnel

- Thermal and material scientists are crucial for understanding and modelling the thermal behaviour of the satellite and selecting materials that can withstand space conditions.
- 1 scientist in Year 1, reduced to 0 in Year 2 as initial research is completed.

Engineering Personnel

- Engineers design, optimise the louver system and ensure the design meets space flight requirements, systems engineers integrate the louver system with other satellite systems, and electrical engineers manage the electronics and control systems.
- 1 engineer in Year 1, increased to 2 in Year 2 for testing and final design.

Technicians

- A technician would build and test prototypes and conduct thermal and mechanical tests.
- 1 technicians in Year 1 and Year 2 for testing and integration.

Administration Personnel

- A project managers would oversee the project timeline, budget, and coordination between teams and maintain detailed records of designs, tests, and modifications.
- 1 administration personnel in Year 1 and Year 2 for additional documentation and coordination.

Management Personnel

- A management personnel would ensure the project aligns with the client's overall mission, objectives, and ensure all components meet the required standards and specifications.

- 1 management personnel in both Year 1 and Year 2.

The personnel costs are outlined in the table below, assuming the inflation rate is 3.6%.

Year	Year 1	Year 2	Cumulative Total
PERSONNEL			
Science Personnel	$90,000	-	$90,000
Engineering Personnel	$100,000	$207,200	$307,200
Technicians	$75,000	$77,700	$152,700
Administration Personnel	$65,000	$67,340	$132,340
Project Management	$135,000	$139,860	$274,860
Personnel Margin	$46,500.00	$49,210.00	$95710
Total Salaries	$511,500	$541,310	**$1,052,810**
Total SuperAnnuation	$86,955	$92,023	**$178,978**
TOTAL PERSONNEL	$598,455	$633,333	**$1,231,788**

4.3 TRAVEL

4.3.1 TOTAL FLIGHT/TRAIN COSTS

Assumptions:

1. Number of Trips: 1 engineering personnel, 1 management personnel, 1 administration personnel will make a 1 domestic trip, and 1 international trip in Year 1, 2 engineering personnel, 1 management personnel, 1 administration personnel will make a 1 domestic trip, and 2 international trip in Year 2.
2. Travel Destinations:
   - Domestic trips within Australia (including but not limited to Sydney to Canberra, Melbourne, or Brisbane).
   - International trips to major space industry hubs (including but not limited to the United States, Europe).
3. Travel Costs:
   - Domestic Flights: Average cost of $100 per round trip.
   - International Flights: Average cost of $500 per round trip.

4.3.2 TOTAL ACCOMMODATION COST

Assumptions

1. Duration of Stay: Each trip will last an average of 2 days for domestic trips, and 3 days for international trips.
2. Hotel Costs:
   - Domestic Hotels: Average cost of $150 per night.
   - International Hotels: Average cost of $250 per night.

4.3.3 TOTAL TRANSPORTATION COST

Assumptions

1. Transportation Mode:
   - Domestic trips: Rental cars or taxis.
   - International trips: Rental cars, taxis, or shuttle services.
2. Transportation Costs:
   - Domestic Transportation: Average cost of $30 per trip (round trip from airport to hotel).
   - International Transportation: Average cost of $30 per trip (round trip from airport to hotel).

4.3.4 TOTAL PER DIEM COST

Assumptions

1. Per Diem Rates:
   - Domestic Per Diem: $20 per day (covering meals and incidental expenses).
   - International Per Diem: $30 per day (covering meals and incidental expenses).

4.3.5 TRAVEL MARGIN

1. Budgeted Travel Costs: Assume the budgeted travel costs are 10% higher than the actual travel costs.

The summary of the travel costs is outlined below:

TRAVEL			
Total Flights/Train Cost	$1,800.00	$4,400.00	**$ 6,200**
Total Accommodation Cost	$3,150.00	$7,200.00	**$ 10,350**
Total Transportation Cost	$180.00	$360.00	**$ 540**
Total Per Diem Cost	$390.00	$880.00	**$ 1,270**
Travel Margin	$552.00	$1,284.00	**$ 1,836**
Total Travel Costs	$ 6,072	$ 14,632	**$ 20,704**

4.4 DIRECT COSTS

Estimated costs for the mechanical subsystem, power subsystem, thermal control subsystem, manufacturing facility, test facility, and facility cost margin for the team in both Year 1 and Year 2:

DIRECT COSTS			
Mechanical Subsystem	$3,800.00	$5,000.00	$ 8,800
Power Subsystem	$2,000.00	$3,000.00	$ 5,000
Thermal Control Subsystem	$1,000.00	$1,500.00	$ 2,500
[redacted]			
Total Spacecraft Direct Costs	$ 6,800	$ 9,842	$ 16,642
Manufacturing Facility Cost	$2,000.00	$2,500.00	$ 4,500
Test Facility Cost	$1,500.00	$2,000.00	$ 3,500
Facility Cost Margin	$1,750.00	$2,250.00	$ 4,000
Total Facilities Costs	$ 5,250	$ 6,993	$ 12,243
Total Direct Costs	$ 12,050	$ 16,835	$ 28,885
Total MTDC	$ 6,800	$ 9,842	$ 16,642
Total F&A	$ 680	$ 984	$ 1,664

4.5 SUMMARY OF THE ESTIMATED BUDGET FOR THE SPACE MISSION

FINAL COST CALCULATIONS			
Total Projected Cost	$ 617,257	$ 17,819	$ 635,076
Total Cost Margin	$ 48,802 7.9%	$ 52,744 296.0%	$ 101,546
Total Project Cost	$ 666,059	$ 70,563	$ 736,622

By considering all the components and their associated costs, the total project cost for the mission is estimated to be $736,622.